# ***Task***

**1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.**

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**Steps**

1. Place 1 Server, 1 Switch, and 2 PCs.
2. Connect all devices using Copper Straight-Through cables.
3. Configure the server:
    - IP Address: 192.168.1.2 ○ Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
4. Open Server → Services → DHCP.
5. Turn DHCP Service ON.
6. Create a DHCP Pool: ○ Pool Name ○ Default Gateway ○ DNS Server
    - Start IP Address ○ Subnet Mask
    - Maximum Users
7. On both PCs: ○ Desktop → IP Configuration → DHCP.

Example Result

| Device | Assigned IP |
|---|---|
| PC0 | 10.0.0.1 |
| PC1 | 10.0.0.2 |
| PC2v | 10.0.0.3 |

Both PCs receive different IP addresses automatically from the DHCP pool.

**2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.<br><br><em><strong>Hint:</strong> Use arrows to show the direction of each message between client and server.</em>**

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**DHCP Process**

```
+------------------+              +-------------------+
|   DHCP Client   |              |   DHCP Server    |
+------------------+              +-------------------+

         |                                  |
         |-------- DHCP Discover ------------------------>|
         |     (Client searches for DHCP Server)       |
         |                                  |
         |<------- DHCP Offer -----------------------------|
         |   (Server offers an available IP Address)    |
         |                                  |
         |-------- DHCP Request ------------------------->|
         | (Client requests the offered IP Address)     |
         |                                  |
         |<------- DHCP Acknowledge (ACK) ---------------|
         | (Server confirms and assigns the IP Address)  |
         |                                  |
```

Steps

- Discover: Client searches for a DHCP server.
- Offer: Server offers an available IP address.
- Request: Client requests the offered IP.
- Acknowledge: Server confirms and assigns the IP.

**3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**

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Configuration

- Start IP: 192.168.1.100
- Maximum Users: 1

**Result**

| Device | IP Address |
|---|---|
| PC0 | 192.168.1.100 |
| PC1 | No IP Assigned (DHCP Failed) |

**Explanation**

Only one IP address is available in the DHCP pool. The first PC receives the IP address, while the second PC cannot get one because there are no free addresses left.

**4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

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```
          Internet
              |
         Wi-Fi Router
              |
         DHCP Server
           /     \
      Laptop    Smartphone
```

**DHCP Configuration**

- Pool Start IP: 192.168.10.100
- Maximum Users: 20

**Assigned IPs**

| **Device** | **Assigned IP** |
|---|---|
| Laptop | 192.168.10.100 |
| Smartphone (PC) | 192.168.10.101 |

**Why DHCP is Useful**

DHCP automatically assigns IP addresses to customers' devices when they connect to the restaurant's Wi-Fi. This avoids manual configuration, saves time, and prevents IP address conflicts.